

Investment Portfolios

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1 Motivation

A common piece of investing advice goes as follows:¹

Stock-based funds have higher long-term return potential but also have relatively high volatility. On the other hand, bond funds are more stable, especially in terms of income, but don't have the long-term growth potential of stocks. Therefore, it's generally suggested that younger investors put most of their money into stocks and gradually shift their investments into bonds as they get closer to retirement.

For investors that don't want to actively manage their retirement portfolio, companies such as Vanguard have introduced “Target Date Funds” (TDFs), which are marketed as a “one stop shop” for retirement investing. [Vanguard's explanation](#) states, “managers gradually shift each fund's asset allocation to fewer stocks and more bonds so the fund becomes more conservative the closer you get to retirement” (see Figure 1). My employer plan goes straight to a TDF, and I am not alone: according to this [Forbes article](#), most employer-sponsored retirement funds default to TDFs, and it is predicted that 88% of retirement plan contributions will flow into TDFs by 2019.

TDFs are popular, but does their approach actually make sense? In this post, I investigate that question, taking as given the assertion that stocks are riskier than bonds, but have a higher expected return. I start by introducing a simple model that will frame my thinking, present cases against and for shifting assets from stocks to bonds over time, and close with an informal discussion.

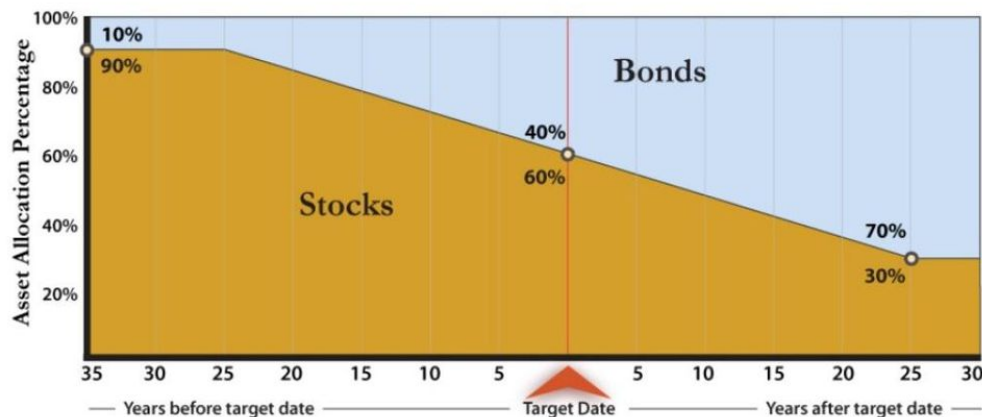


Figure 1: Mix of stocks and bonds in a Target Date Fund over time. Source: US Department of Labor [investment bulletin](#) on TDFs.

¹Quote taken from [the Motley Fool](#).

2 Model

An investor starts with \$1. In each period t , she may place a fraction α_t of her wealth in a risky asset, with (unknown) return r_t ; the remaining $1 - \alpha_t$ fraction of her wealth goes into a safe asset with known return s . Given returns r_t and investment decisions α_t , her per-period wealth W_t evolves as follows:

$$W_{t+1} = (\alpha_t r_t + (1 - \alpha_t)s)W_t.$$

Assume that $s \geq 1$ and r_t are iid draws from a known distribution G , with $E[r_t] > s$ and $P(r_t < s) > 0$. Our investor plans to retire in period T . Her final utility is given by $U(W_T)$ for some known and weakly increasing function U , and she invests to maximize her expected utility.

3 The Contrarian

I start by presenting evidence *against* the conventional wisdom that investors should gradually shift from stocks to bonds.

Proposition 1 *If investors maximize expected wealth ($U(x) = x$), then it is optimal to select $\alpha_t = 1$ for all t .*

Of course, most investors don't aim to maximize expected wealth. Generally, money is assumed to have diminishing marginal returns. One standard assumption that captures this effect is to take $U(x) = \log(x)$, implying that in order to increase the utility of two different investors by equal amounts, it is necessary to increase their wealth by equal *percentages*. In this case, it may make sense to invest in a mix of bonds and stocks, but the optimal mix does not depend on one's wealth level or the time until retirement.

Proposition 2 *If investors maximize the expected logarithm of wealth ($U(x) = \log(x)$), then there exists $\alpha \in [0, 1]$ such that it is optimal to select $\alpha_t = \alpha$ for all t .*

Of course, $U(x) = \log(x)$ is still one particular functional form. However, it turns out that for *any* utility function U , committing to any particular trajectory of the stock-bond mix is no better than committing to the reverse.

Proposition 3 *If investors follow a non-adaptive policy² $\alpha = (\alpha_1, \dots, \alpha_{T-1})$, then the distribution of W_T is invariant to a permutation in α .*

²I say that a policy is non-adaptive if α_t is independent of the vector of returns $\mathbf{r}$.

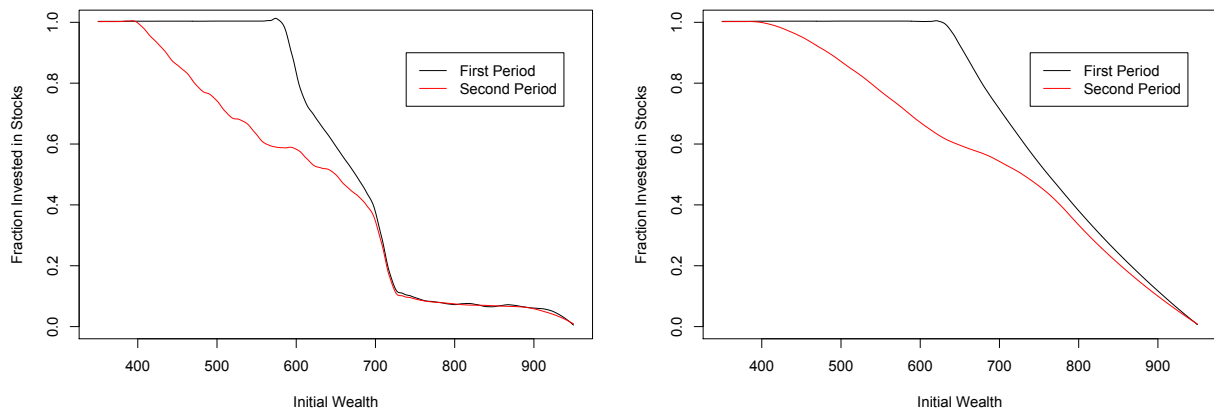


Figure 2: Average percentage of portfolio invested in stocks under optimal policy, as a function of initial wealth. Left side is $U(x) = (1 + e^{-(x-b)/50})^{-1}$, right side is $U(x) = \frac{1}{2}\sqrt{x/650}$.

4 Adaptive Investment

Proposition 3 establishes that among non-adaptive investment policies, there is no reason that the investment in stocks should decrease over time. However, in practice investors can use adaptive policies. For example, I might shift my money to bonds if I know I have enough to retire on, but keep it in stocks if I had bad initial luck and need to hope for big returns in the future.

Given any utility function U and distribution of returns G , it is possible to use dynamic programming to solve for the optimal adaptive policy (as a function of the current wealth level and time until retirement). The results are given in Figure 2. (Code provided in Section 6 – no apologies for its limitations, as one of the joys of being an academic is being able to write non-robust code.)

My intuition for these plots is that the desire to *acquire information* about returns encourages the investor to front-load her investment in stocks. This is illustrated by the simple example below.³

Example 1 Suppose that $s = 1.1$ and $r \in \{L, M, H\}$, with $L = 0.8, M = 1.2, H = 1.65$. Let $p_L = P(r_t = L)$, and similarly for p_M and p_H . Suppose that an investor has \$500,000 in savings, is two years from retirement, and wants to retire with at least \$650,000 : $U(w) = \mathbf{1}(w \geq \$650,000)$.

Let us first consider non-adaptive strategies. The investor cannot invest in safe assets alone, as she will not reach her retirement goal. If she invests only in risky assets, she will succeed so long as neither period has low returns, or at least one period has high returns. Her chance of success is $1 - p_L(p_L + 2p_M)$. If she invests in safe assets in the first period, and risky assets in the second, she will succeed whenever the return in the second period is not L . Her success probability is $1 - p_L$.

An optimal strategy is adaptive. By investing in risky assets in the first period, the investor gains information. If her initial return is M or H , she can lock her wealth in safe assets during the second period; if her initial return is L , she knows that she must invest in risky assets in the second period, hoping for a high return. Her probability of success under this strategy is $1 - p_L(p_L + p_M)$, and on average her portfolio has more safe assets in the second period.

³I would be curious establish conditions under which the investor expects to hold a higher percentage of stocks in the first period than the second. One tricky aspect is that while the value function is unique, the optimal policy may not be. I know that there sometimes exist optimal strategies under which the expected fraction of the portfolio invested in stocks increases over time, but it is conceivable to me that there always exists an optimal strategy under which the investor expects to gradually shift to bonds.

5 Discussion

So, does the conventional wisdom make sense? My answer is a qualified yes. An investor who actively manages her portfolio to prepare for retirement might reasonably expect to see the percentage of stocks decline over the years. The reason for this is that investing in stocks early *provides information* which can be used to make future decisions⁴ – if this information is not used, then there is no reason to gradually shift the portfolio from stocks to bonds.

This suggests that sales pitch for Target Date Funds (TDFs) doesn't really make sense – they cater to investors who want to set aside money and forget about it until retirement, but adopt a “glide” strategy that can't easily be justified for these investors.

As one [Investopedia article](#) states,

One retiree may have enough money on hand to invest strictly in bonds and other fixed-income securities. Another, requiring both growth and income, may need an equity component to keep the portfolio on track. A fund that meets the needs of one of these investors is unlikely to meet the needs of the other.

Will I redirect my contributions away from the Target Date Fund? Not right away – it's mostly stocks right now, as it likely should be. But don't be surprised if I end up shifting the money more actively at some point in the future.

6 Code

```
#Given vectors X,Y such that X is increasing and Y = g(X),
#returns a discretized function approximating g
approximate = function(X,Y){
  return(Vectorize(function(x){
    i = max(c(which(X < x),1))
    return(Y[i]) #Could do linear interpolation, but why bother?
  })))
}

#Given current wealth w, return from safe and risky assets, and value function V,
#computes expected utility from any given alpha
#Assumes p is probability vector of the same length as r, and s and w are scalars
expectedUtility = function(w,s,r,p,V){
  return(Vectorize(function(alpha){
    return(sum(V((alpha*r+(1-alpha)*s)*w)*p))
  })))
}

s = 1 #return from safe asset
r = .6 + c(0:10)/10 #possible returns from risky asset
p = rep(1/length(r),length(r)) #p[i] is probability of return r[i]

a = 0.02; b= 650;
N = 250
W = b+6*rep(-N:N)/(a*N) #Different levels of wealth, in a range relevant to U
```

⁴In the model, the only decision is the mix of stocks and bonds, but in reality an investor may also choose to change consumption behavior before retirement or work for a few extra years.

```

#U for plot 1
U = Vectorize(function(x){return(1/(1+exp(-a*(x-b))))})

#U for plot 2
U = Vectorize(function(x){return(sqrt(x/b)/2)})

#Target of 650
#U = Vectorize(function(x){if(x>=650) return(1); return(0)})

T = 2

#V[t+1,i] gives value of having W[i] dollars with t periods remaining
V = matrix(0,nrow = T+1,ncol=length(W))
#Alpha[t+1,i] gives optimal portfolio wit W[i] dollars and t periods remaining
Alpha = matrix(0,nrow = T+1,ncol=length(W))
V[1,] = U(W)
for(t in 1:T){
  for(i in 1:length(W)){
    #soln = optimize(expectedUtility(W[i],s,r,p,approximate(W,V[t,])), interval=c(0,1),
    #Alpha[t+1,i] = soln$maximum #Optimal alpha, given t remaining periods and wealth
    #V[t+1,i] = soln$objective #Expected value, given t remaining periods and wealth

    #Optimize function works poorly for locally flat functions,
    #so taking this approach instead
    EU = expectedUtility(W[i],s,r,p,approximate(W,V[t,]))(c(0:N)/N)
    V[t+1,i] = max(EU)
    Alpha[t+1,i] = which(EU==max(EU))[1]/N
  }
}
#filled.contour(V,x = c(0:T),y=W,xlab='Time Remaining',ylab='Wealth')

#Two period model — compare alpha in first period to average alpha in second period
AlphaSecondPdAvg = rep(0,length(W)) #AlphaSecondPdAvg[i] gives average alpha in second period, w
alphafn = approximate(W,Alpha[2,]) #alphafn(w) says what alpha to choose, given wealth of w and d
for(i in 1:length(W)){
  AlphaSecondPdAvg[i] = sum(alphafn((Alpha[3,i]*r+(1-Alpha[3,i])*s)*W[i])*p) #Alpha[3,i] is
}

plot(NULL,xlab='Initial Wealth',xlim=c(W[1],W[length(W)]),ylab='Fraction Invested in Stocks',ylim=
lines(W,predict(loess(Alpha[3,]~W,span=0.1)),col='black')
lines(W,predict(loess(AlphaSecondPdAvg~W,span=0.1)),col='red')
legend(x=W[N*1.3],y=0.9,legend = c('First Period','Second Period'),col=c('black','red'),lty=c(1,1))

```